

Fix or Fake? How Creators Negotiate Cultural Bias in Generative AI Heritage Creation

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Abstract

Generative AI is increasingly adopted in cultural heritage creation, yet its outputs often exhibit cultural bias that challenges creators' judgment and responsibility. Existing studies have largely examined such bias from technical or ethical perspectives, leaving creators' situated practices underexplored. Drawing on a Dunhuang-themed AIGC video creation workshop, this study investigates how creators perceive and negotiate AI cultural bias during human-AI co-creation. We find that bias is not only manifested as cross-cultural Westernization, but also as the loss of intra-cultural granularity within seemingly coherent cultural frames. Creators actively engage in negotiation with AI outputs, balancing cultural accuracy, aesthetic goals, and creative efficiency. We argue that AI cultural bias should be understood as an interactional and practice-based phenomenon, and highlight the need for cultural heritage AIGC systems that support creators' cultural judgment rather than generation alone.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Generative AI, Cultural Bias, Human-AI Co-creation, Cultural Heritage

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1 Introduction

Generative artificial intelligence (AIGC) is increasingly used in the restoration [24, 25], reinterpretation [15], and dissemination [9, 21] of cultural heritage, but its application also exposes persistent cultural bias [6]. Because cultural heritage data are shaped by historical power structures, such biases may be inherited and amplified by generative models, leading to stereotypical or decontextualized representations [26].

Creators act not only as tool users but also as evaluators and final decision-makers, often being the first to perceive and address cultural bias in AI-generated outputs [12]. Prior research has largely examined AI cultural bias from technical or macro-ethical perspectives, leaving creators' situated practices underexplored. In particular, how creators perceive and negotiate AI cultural bias through human-AI collaboration, and how this process reshapes creative practice and cultural judgment, remains insufficiently understood.

To address this gap, we investigate a Dunhuang-themed AIGC video creation workshop through process observation and semi-structured interviews, examining how AI cultural bias manifests in generative creation. This study addresses the following research questions:

RQ1: How do creators perceive, identify, and interpret AI cultural bias in the generative creation of cultural heritage content?

RQ2: How do creators negotiate AI cultural bias through human-AI collaboration, and how does this negotiation reshape their

creative practice and understanding of cultural heritage authenticity and responsibility?

2 Related Work

2.1 AI Cultural Bias and Credibility in Cultural Heritage AIGC

As generative AI becomes embedded in visual art production, scholars increasingly conceptualize it as a sociotechnical system with inherent value orientations [3]. Because training datasets are dominated by Western mainstream cultures, AIGC systems tend to privilege dominant cultural frameworks while marginalizing or de-contextualizing non-Western and minority cultures [23]. Such bias manifests not only through explicit stereotypes but also through implicit cultural homogenization. Even when prompts specify non-Western cultural attributes, generative models often override them through Western symbolic conventions, resulting in semantic distortion and symbolic violence against source cultures [5].

When applied to cultural heritage contexts, these representational biases raise more acute concerns regarding credibility and authenticity [10, 20]. AIGC has evolved from a tool for digital restoration into a medium for interpretation and reinterpretation [4], lowering creative barriers while diversifying digital dissemination [8, 22]. However, unlike general artistic creation, cultural heritage production requires not only visual plausibility [17] but also the accurate transmission of cultural integrity and symbolic meaning [7]. Existing studies reveal a persistent misalignment between the technical logic of AIGC and the value logic of cultural heritage: lacking deep contextual understanding, AI models often reduce heritage to surface-level visual features, leading to arbitrary assemblages or misplacements of cultural symbols. Although such outputs may appear visually persuasive, they risk undermining the credibility of digital heritage narratives, particularly in underrepresented regions such as Latin America and Southeast Asia [23].

2.2 Human-AI Artistic Co-creation

AIGC-based artistic creation is a dynamic process of human-AI collaboration. While prior work highlights AI's potential to expand creative space and support ideation [11, 22], this potential becomes problematic in cultural heritage contexts, where the boundary between serendipitous discovery and cultural offense is highly ambiguous. For heritage practitioners, AI-generated uncertainty often functions less as inspiration than as cognitive noise requiring correction [16].

As a result, human-AI co-creation evolves into ongoing boundary work, in which creators continuously arbitrate between generative flexibility and cultural factuality. Although human-in-the-loop principles emphasize user control [1, 18], how creators actually make sense of cultural bias in situ, and how they distinguish acceptable artistic imagination from non-negotiable cultural inaccuracies in practice, remains insufficiently examined.

3 Methods

To examine creators' micro-level negotiations with AI bias in a high-fidelity creative setting, we conducted a one-week AIGC video creation workshop using a Research through Design approach [19,

27]. This method enabled close observation of how creators balance aesthetic goals, efficiency, and cultural authenticity during situated practice (see Fig. 1).



Figure 1: Procedure of study

3.1 Participants

We recruited 25 participants with art and design backgrounds, organized into seven groups of three to four members each. Group-based work allowed us to observe how cultural judgments and bias negotiations emerged through discussion rather than purely individual decision-making. All participants had prior experience in visual art creation; over 80% had used AIGC tools such as Midjourney, Runway, or Stable Diffusion, and 50% had publicly released mature AIGC works. This expert-oriented sample ensured that participants were capable of identifying tool-related bias rather than being limited by technical unfamiliarity.

3.2 Tasks and Procedure

The workshop focused on Dunhuang culture as the creative theme, selected for its distinctive Eastern aesthetics and strong historical specificity, making it suitable for examining AI's cross-cultural understanding and hallucination-related failures. A non-interventionist observation strategy was adopted, and the workshop consisted of four stages:

- **Introduction:** Introduced core Dunhuang visual elements and the AIGC video production workflow to establish a shared domain and technical baseline.
- **Creation:** Each group created a 3–5 minute narrative video using AIGC tools, with no stylistic constraints.
- **Presentation:** Groups presented final works and creative materials (e.g., prompt logs, iteration histories), reflecting on creative intent and cultural conflicts.
- **Individual interviews:** Conducted one-on-one semi-structured interviews (1,186 minutes total) using stimulated recall to examine decision-making at critical moments.

3.3 Data Collection and Analysis

The data corpus comprised: (1) demographic questionnaires; (2) process artifacts, including prompt logs, failed intermediate outputs, final videos, and creative documentation; (3) audio recordings and transcripts from interviews and presentations; and (4) observational field notes. All participants provided informed consent prior to data collection.

Qualitative data were analyzed using Braun and Clarke's thematic analysis [2]. Two researchers independently conducted open coding, focusing on bias identification, emotional responses, and

coping strategies. Coding discrepancies were resolved through regular consensus meetings, resulting in four core themes and ten subthemes.



Figure 2: Workshop: (a) Introduction; (b) Presentation; (c) Output

4 Results

4.1 Workshop Outcomes

All seven groups completed creative adaptations and short video productions centered on Dunhuang culture using AIGC tools, and presented their creative concepts, processes, and reflections during the workshop. Figure 2-c shows representative cover images of the final works. The outputs spanned multiple formats, including animated shorts, experimental videos, and mixed-style visual compositions.

4.2 Interview Findings

4.2.1 Manifestations of Cultural Misalignment. We conceptualize cultural bias as a structural condition embedded in generative models, which becomes perceptible to creators through concrete instances of misalignment during co-creation. For textual outputs, creators relied on common sense for verification, whereas cultural bias in image and video generation was more subtle and complex. Three major forms of cultural misalignment were identified.

Westernization and representational homogenization were the most salient cross-cultural biases. Models frequently reconstructed Eastern concepts using Western visual symbols, resulting in semantic substitution. For example, P1 noted that a prompt for a “Western Regions of China”-style temple produced a Western-style building, indicating a failure to distinguish “the West” as a geographic versus cultural construct. Asian faces also exhibited mode collapse with limited diversity (P20).

Loss of intra-cultural granularity was a key finding. Bias occurred not only across cultures but also within a single culture, flattening differences across regions and historical periods. AI systems lacked fine-grained understanding of specific dynasties and sites, leading to temporal and spatial mismatches, for example, confusing grotto typologies unique to Longmen (P18) or blurring stylistic distinctions between the Tang and Yuan dynasties (P21). This generalized Orientalism homogenized rich historical detail.

Data-scarcity-driven structural hallucinations were also common. Creators reported malformed structures and abnormal proportions, such as extra fingers in Buddhist hand gestures (P20). These errors reflected low-resource cultural domains, where insufficient training data led to plausible but incorrect structural outputs.

4.2.2 How Creators Verified Cultural Bias in AIGC Outputs. To address these biases, creators assumed the invisible labor of acting as fact-checkers, revealing a dynamic and conditional trust in AIGC systems.

When lacking clear judgment criteria, creators attempted **model triangulation**, comparing outputs across multiple AI tools to assess credibility. As P7 described, “Using two AI tools to verify each other, using one black box to test another.” This strategy reflected an attempt to regain psychological security at the output level in the absence of interpretability.

For culturally critical elements, creators **returned to traditional authorities**, consulting historical records, books, and web sources for ground-truth calibration.

A tension between **cognitive confidence and anxiety** also emerged. Creators with stronger domain knowledge tended to rely on personal judgment (e.g., P3’s reasoning about Silk Road trade scenes), whereas others expressed concern about their limited ability to detect fabricated outputs. As P12 noted, “many people know very little about cultural heritage... even if the AI fabricates information, creators may not be able to tell.” This reveals a paradox of cultural heritage AIGC, in which lowered creative barriers coexist with heightened epistemic demands.

4.2.3 Human-AI Negotiation Strategies: Balancing Authenticity and Workload. Creators’ responses to identified biases varied according to perceived repair costs and creative goals, resulting in four negotiation strategies.

Resistance and repair was the most common approach. Driven by a sense of responsibility toward cultural authenticity, creators repeatedly refined prompts, trained LoRA models, or manually edited outputs. As P16 stated, “historical facts must be respected,” while P15 expressed frustration with the repair process: “it doesn’t understand, it won’t change, so in the end I had to fix it myself.”

Pragmatic satisficing emerged when repair costs became excessive or deadlines approached. P1’s remark, “You kind of deceive yourself... if it works, just use it,” reflects acceptance of a suboptimal but usable output. This form of learned helplessness risks normalizing cultural bias through creative fatigue.

Aesthetic negotiation involved balancing artistic effect and authenticity. Some creators reframed AI errors as inspiration while clearly distinguishing algorithmic mistakes from intentional artistic choices. As P2 emphasized, “this is not an intentional artistic decision; it’s simply a mistake.”

Finally, **strategic avoidance** was more common among experienced creators, who filtered themes in advance or abandoned AI tools altogether to prevent cultural conflicts.

5 Discussion

This study examines how creators identify and negotiate AI cultural bias in cultural heritage-oriented generative creation, and how this process reshapes creative practice. Our findings show that such bias manifests in concrete ways that shape creators’ judgments and workflows.

5.1 How AI Cultural Bias Is Perceived and Identified in Cultural Heritage Creation

Prior research has focused on AI’s macro-level neglect of marginalized cultures [13]. Our findings resonate with this line of work: creators in our study similarly encountered Westernization and cultural homogenization in AI-generated heritage content. However, by examining these issues at the micro-interaction level, our findings show that bias in cultural heritage creation is characterized not simply by incorrectness but by a loss of cultural granularity. Fine-grained mismatches (e.g., conflating Tang and Yuan dynasties or Dunhuang and Longmen) indicate a tendency to compress rich historical distinctions into generalized and stereotypical “Orientalist” representations.

This form of intra-cultural bias is more concealed than cross-cultural bias and reflects models’ limited capacity for context-specific cultural knowledge. Bias recognition thus becomes a process through which creators actively counter decontextualization by reintroducing cultural granularity through their own interpretive judgment.

5.2 Credibility Judgment and Invisible Labor in Cultural Heritage AIGC

In cultural heritage AIGC research, historical authenticity and cultural context are often treated as stable evaluative principles [7, 17]. Our findings indicate that in situated creative practice, credibility cannot be operationalized as fixed criteria but is instead dynamically constructed through creators’ invisible labor.

Rather than simply labeling outputs as “accurate” or “distorted,” creators negotiate among cultural responsibility, aesthetic expression, production efficiency, and communicative impact. Depending on contextual constraints, they may resist bias, accept limited distortion, or consciously compromise. When AI systems fail to offer cultural justification, epistemic responsibility shifts to human creators, who must arbitrate among competing values.

This reveals an automation paradox in cultural heritage AIGC: while such tools aim to lower creative barriers, they raise the level of prior knowledge required to avoid cultural misinterpretation. Creators with limited historical expertise are thus more vulnerable to confidently fabricated AI outputs. Without adequate verification mechanisms, AIGC risks reinforcing disparities among creators rather than enabling inclusive cultural creation.

5.3 Human-AI “Cultural Negotiation” as a Creative Practice Mechanism

Prior research emphasizes human agency in co-creation [14]. Our findings resonate with this perspective: participants actively engaged in repeated prompt refinement, manual correction, LoRA training, and strategic filtering to restore cultural accuracy. At the same time, we show that such agency is negotiated and labor-intensive under biased model conditions rather than effortless or fully autonomous. Under persistent bias-related friction and time constraints, agency becomes fragile, giving way to compromise or avoidance. Creators’ negotiation with AI thus reflects a satisficing logic shaped by constrained cognitive and temporal resources.

This dynamic introduces an underexplored risk of bias laundering: when creators opt for aesthetic compromise under time

pressure or creative temptation, AI-generated bias may receive implicit human endorsement and circulate as human-authored work. Without low-cost correction mechanisms, such negotiation can normalize and stabilize cultural bias.

5.4 Design Implications: From Generative Tools to Cultural Meaning-Making Systems

Our findings show that in cultural heritage AIGC creation, responsibility for cultural judgment is largely shifted to creators, manifesting as invisible labor that raises creative barriers and amplifies inequality in bias recognition and correction.

Accordingly, we argue that cultural heritage AIGC should move beyond optimizing generative performance toward supporting cultural meaning-making. Systems should provide cultural traceability by indicating possible sources or references of generated elements to reduce verification costs, as well as fine-grained controls over historical periods or regional styles to enable localized correction. Moreover, fluency should not always be maximized: for culturally sensitive symbols, systems should introduce intentional interactional friction (e.g., confirmation prompts) to discourage unreflective acceptance of cultural bias.

6 Limitations and Future Works

Prior work suggests that creators’ familiarity with a cultural heritage domain may shape their acceptance and modification of AI-generated content. This study did not quantitatively differentiate participants based on their knowledge of Dunhuang culture. Future research could compare creators with different levels of cultural expertise to examine its impact on bias perception and negotiation strategies.

In addition, our design implications are derived from observations of existing commercial AIGC tools and have not been validated through dedicated prototypes. Future work may design and evaluate AIGC systems tailored for cultural heritage creation to assess how design interventions support cultural judgment and reduce creative burden.

7 Conclusion

Based on a Dunhuang-themed AIGC video creation workshop, this study demonstrates that AI cultural bias in heritage contexts is not merely a technical flaw addressable through model improvement, but a practical issue emerging through creators’ interaction with generative systems. Creators identify bias through concrete generation anomalies and respond via verification and adjustment, negotiating among aesthetics, communication goals, and cultural responsibility. As a result, credibility in cultural heritage AIGC is not a fixed ground truth but an acceptable threshold shaped by real-world constraints.

More broadly, our findings reveal an automation paradox: when generative systems fail to capture cultural granularity, responsibility for cultural judgment is shifted back to human creators as invisible labor. This shift disproportionately risks creators with limited cultural expertise, as algorithmic bias may be unknowingly internalized. We therefore argue that cultural heritage AIGC should move

from replacing creation toward supporting judgment, enabling creators to identify uncertainty, define controllable boundaries, and intervene and reinterpret outputs in informed ways.

References

- [1] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N Bennett, Kori Inkpen, et al. 2019. Guidelines for human-AI interaction. In *Proceedings of the 2019 chi conference on human factors in computing systems*. 1–13.
- [2] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.
- [3] Aylin Caliskan, Joanna J Bryson, and Arvind Narayanan. 2017. Semantics derived automatically from language corpora contain human-like biases. *Science* 356, 6334 (2017), 183–186.
- [4] Francesca Condorelli and Francesca Berti. 2025. Creativity and awareness in co-creation of art using artificial intelligence-based systems in heritage education. *Heritage* 8, 5 (2025), 157.
- [5] Fabio De Almeida and Sónia Rafael. 2024. Bias by Default: Neocolonial Visual Vocabularies in AI Image Generating Design Practices. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–8.
- [6] Anna Foka and Gabriele Griffin. 2024. AI, cultural heritage, and bias: Some key queries that arise from the use of GenAI. *Heritage* 7, 11 (2024), 6125–6136.
- [7] Clifford Geertz. 1973. The interpretation of cultures New York. NY: *Basic Books* (1973).
- [8] Shiwen Lai, Yihuan Tian, and Qingfeng Zhang. 2025. The impact of AI-generated technologies-driven digital cultural heritage platforms on users' offline cultural participation intentions. *npj Heritage Science* 13, 1 (2025), 574.
- [9] Jingke Li, Guoyu Sun, Congyun Tang, Wenjuan Chen, Wenwen Yang, Wenxuan Kou, Zhonghe Ruan, Wanjing Ma, and Xuran Nie. 2024. Silk Road journey: A real-time AI-based interactive art installation for Silk Road cultural reenactment and experience. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–5.
- [10] Ruoxi Liu and Haoran Qiu. 2025. Collaboration with GAI in Artistic Creation of Cultural Heritage: Approaches, Challenges and Mitigations. *Proceedings of the Association for Information Science and Technology* 62, 1 (2025), 1562–1564.
- [11] Katherine O'Toole and Emőke-Ágnes Horvát. 2024. Extending human creativity with AI. *Journal of Creativity* 34, 2 (2024), 100080.
- [12] Srishti Palani and Gonzalo Ramos. 2024. Evolving roles and workflows of creative practitioners in the age of generative AI. In *Proceedings of the 16th Conference on Creativity & Cognition*. 170–184.
- [13] Uwe Peters and Mary Carman. 2024. Cultural bias in explainable AI research: A systematic analysis. *Journal of Artificial Intelligence Research* 79 (2024), 971–1000.
- [14] S Shyam Sundar. 2020. Rise of machine agency: A framework for studying the psychology of human-AI interaction (HAI). *Journal of computer-mediated communication* 25, 1 (2020), 74–88.
- [15] Ye Tao, Xiaohui Fu, Jiaying Wu, Ze Bian, Aiyu Zhu, Qi Bao, Weiyue Zheng, Yubo Wang, Bin Zhu, Cheng Yang, et al. 2025. AIFiligree: A Generative AI Framework for Designing Exquisite Filigree Artworks. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–18.
- [16] Simona Tiribelli, Sofia Pansoni, Emanuele Frontoni, and Benedetta Giovanola. 2024. Ethics of artificial intelligence for cultural heritage: Opportunities and challenges. *IEEE Transactions on Technology and Society* 5, 3 (2024), 293–305.
- [17] Chuyao Wang and Nor Azura Adzharuddin. 2025. Synthetic Authenticity and Audience Trust in AI-Generated Intangible Cultural Heritage: A Qualitative Multimodal Study of Chinese Digital Heritage Platforms. *e-Journal of Media and Society* 8, 2 (2025), 1–10.
- [18] Dakuo Wang, Elizabeth Churchill, Pattie Maes, Xiangmin Fan, Ben Shneiderman, Yuanchun Shi, and Qianying Wang. 2020. From human-human collaboration to Human-AI collaboration: Designing AI systems that can work together with people. In *Extended abstracts of the 2020 CHI conference on human factors in computing systems*. 1–6.
- [19] Nan Wang, Hyunsuk Kim, Junfeng Peng, and Jiayi Wang. 2025. Exploring creativity in human-AI co-creation: a comparative study across design experience. *Frontiers in Computer Science* 7 (2025), 1672735.
- [20] Xue Wang and Gino Yu. 2021. Intangible Cultural Heritage and We-Media: The Strategic Design of Chinese Shadow Puppetry Dissemination via Tik Tok. In *Congress of the International Association of Societies of Design Research*. Springer, 1288–1304.
- [21] Wenwen Yang, Wanyi Miao, Guoyu Sun, Yeheng Wang, Changran Zhao, Yilin Zhang, Wenxuan Kou, Yaning Zhang, Xuran Nie, and Jingyu Liu. 2025. Path of Light: Interactive Narrative Design Based on Mix Reality for Silk Road Cultural Perception. In *Adjunct Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*. 1–3.
- [22] Yifan Yang. 2025. AIGC and Human Collaborative Creation Model: A New Paradigm of Digital Art. In *2025 4th International Conference on Science Education and Art Appreciation (SEAA 2025)*. Atlantis Press, 492–499.
- [23] Lili Zhang, Xi Liao, Zaijia Yang, Baihang Gao, Chunjie Wang, Qiuling Yang, and Deshun Li. 2024. Partiality and misconception: Investigating cultural representativeness in text-to-image models. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–25.
- [24] Ying Zhang, Zejian Li, Jiesi Zhang, Fang Hu, Kewen Zhu, Qi Liu, Huanghuang Deng, Xiaoyu Chen, and Lingyun Sun. 2025. Ink Restorer: Virtual Restoration of Ancient Chinese Paintings Inheriting Traditional Restoration Processes. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [25] Shuo Zhao, Yifei Huang, Xiaoyang He, Xin Tong, Xin Li, and Dan Wu. 2025. Reviving Mural Art through Generative AI: A Comparative Study of AI-Generated and Hand-Crafted Recreations. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–20.
- [26] Zhongwen Zhou, Yue Xi, Suchuan Xing, and Yizhe Chen. 2024. Cultural bias mitigation in vision-language models for digital heritage documentation: a comparative analysis of debiasing techniques. *Artificial Intelligence and Machine Learning Review* 5, 3 (2024), 28–40.
- [27] John Zimmerman, Jodi Forlizzi, and Shelley Evenson. 2007. Research through design as a method for interaction design research in HCI. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 493–502.